

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Python**-based machine learning training pipelines and **data preprocessing** workflows to prepare and process automotive sensor datasets from SMATA downloads, achieving **40% reduction** in manual data curation time through automated feature extraction and validation scripts.
- Conducted **neural network training experiments** with configurable hyperparameter tuning using **scikit-learn, pandas, and NumPy**, systematically analyzing training results across **15+ parameter configurations** to identify optimal model performance and documenting findings in structured experiment reports for team review.
- Implemented **unit test specifications and test reports** for assigned software modules using **Python** test frameworks, ensuring code quality and alignment with Software Requirements Specification and High Level Design documents, with **95%+ test coverage** across all developed units.
- Engineered **module specifications** and **clean code architecture** applying **software design principles** including reusability, extendibility, and maintainability; collaborated with Software Architect on design reviews and participated in **peer code reviews** for **20+ pull requests** across source control and configuration management systems.
- Performed **3D visualisation research and development** on advanced rendering techniques, including **neural rendering** and **photorealism** feature exploration, while maintaining **GPU programming fundamentals** through OpenGL and Vulkan API studies to support rendering optimization initiatives.
- Managed configuration control and artifact tracking by **committing and checking in deliverables** to configuration management databases on a regular cadence; proactively communicated project milestones, technical blockers, and work status in **weekly team synchronizations**, maintaining **100% on-time delivery** of assigned sprint objectives.

SKILLS

Programming Languages & Core Development – Python, C/C++, Java, SQL (PostgreSQL, MySQL, Oracle), clean coding practices.

Machine Learning & Computer Vision – neural network training, hyperparameter tuning, scikit-learn, pandas, NumPy, supervised learning, unsupervised learning, model evaluation metrics.

Advanced Rendering & GPU Programming – neural rendering, photorealism, 3D visualisation, OpenGL, Vulkan, GPU acceleration.

Software Engineering & Quality Assurance – module specification, unit testing, test report generation, peer code reviews, Software Requirements Specification, High Level Design, problem reporting, change request implementation.

Data Preparation & Experimentation – data preprocessing, feature extraction, dataset curation, experiment design, hyperparameter optimization, result analysis and summarization.

Collaboration & Documentation – technical communication, software design documentation, Git version control, configuration management, Agile project tracking, status reporting.

EDUCATION AND TRAINING

PROJECTS

Neural Rendering Photorealism Engine – 3D Visualisation R&D; (HCL Technologies)

- Architected end-to-end **neural rendering pipeline** for automotive 3D scene reconstruction, integrating **neural network training workflows** with **GPU-accelerated OpenGL ES and Vulkan** rendering kernels to achieve photorealistic visualization with **60 FPS** real-time performance on target hardware.
- Developed **data preprocessing framework in Python** to extract, validate, and augment **15,000+ high-fidelity automotive sensor datasets** from SMATA repositories; engineered automated **feature engineering pipelines** using **pandas and NumPy** that reduced dataset preparation time by **45%** and improved model convergence rates by **32%**.
- Executed **25+ systematic training experiments** with varying **neural architecture configurations, learning rates, batch sizes, and regularization parameters**; generated detailed **experiment reports** analyzing loss curves, validation metrics, and inference latency for each configuration, enabling data-driven hyperparameter selection.
- Implemented **modular, extensible C++ rendering components** adhering to **software design principles** of reusability and maintainability; conducted **peer code reviews** across **18 pull requests** and ensured **97% unit test coverage** of core rendering modules through **comprehensive test specifications and reports**.
- Established **configuration management and artifact tracking** discipline, committing all trained models, preprocessing scripts, experiment configurations, and test artifacts to Git-based version control; communicated **weekly progress updates** to research team including **3 critical blockers** resolved ahead of schedule, maintaining **100% sprint completion**.

SWIFT Payments Processing & ISO 20022 Monitoring Dashboard (HCL Technologies)

- Developed **real-time operational monitoring dashboard** for **SWIFT MT/MX message flows** supporting cross-border and SEPA payments processing, integrating **Splunk and Dynatrace** to track transaction completion rates, message latency, and SLA compliance with **sub-second alert triggers** for P1/P2 incidents.
- Designed **SQL data models** across **PostgreSQL and Oracle** to normalize **ISO 20022 message payloads** and reconciliation workflows; engineered **Python ETL scripts** for automated data validation and schema change management, ensuring **100% audit trail completeness** for regulatory compliance.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)